

# Keep the Angle

A Universal Geometry-Preserving Basis in Spectral Embeddings

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## Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [11]. We show the geometry does not vanish—it migrates entirely into the *angular* coordinate. Writing each node’s low-mode embedding in polar form (magnitude  $\times$  direction), the direction preserves graph geodesics at Spearman  $\rho \approx 0.89$  on a reference manifold, while the magnitude preserves  $\rho \approx 0.03$  and a random-mode basis of equal size preserves  $\rho \approx 0.04$ . Keeping only the angle is exactly the row-normalization of Ng–Jordan–Weiss spectral clustering [5], whose existing theory concerns cluster separation [9]; we give it a geodesic justification. We show, as a corollary of von Luxburg–Radl–Hein, that the radial coordinate of the full embedding degenerates to a local-degree quantity for intrinsic dimension  $d \geq 2$ , and **conjecture**, with strong empirical support, that the angular coordinate remains rank-faithful to geodesic distance—flat in both mode-count and graph size where the raw embedding decays. The basis is **universal**: it preserves geodesics on any substrate carrying genuine low-dimensional Riemannian geometry—random-geometric lattices, manifold-embedding trajectories, and Wolfram-model hypergraph rewriting alike—and fails correctly on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ( $d \approx 1\text{--}3.6$ ), fidelity falls approximately linearly with intrinsic dimension.

## 1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [2]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [11]. Yet the most successful spectral-clustering algorithm [5] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [9, 10]. It does not address metric geometry. **This paper asks what row-normalization recovers of the graph’s geodesic structure, and why.**

Our answer is a clean decomposition. Write each node’s low-mode embedding in polar form, magnitude  $\times$  direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg’s theorem kills; the *direction* carries the graph’s geodesic geometry. Row-normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

This “keep the scale-invariant direction” operation has analogues beyond clustering—for example polar-coordinate KV-cache quantization, which represents key vectors in magnitude–direction form and quantizes the two parts separately [7]. It also admits a physical reading: in observer theory a bounded observer perceives geometry only after coarse-graining a discrete substrate [14], and the angular projection is a candidate for that coarse-graining. We keep that interpretation to the discussion (§7) and let the spectral result stand on its own; the empirical core is that the basis is *universal*, preserving geodesics on any graph carrying genuine low-dimensional Riemannian geometry, independent of how it was generated.

**Contributions:** (1) the **Keep-the-Angle Principle** (§3)—a Radial-Degeneracy *theorem* (for the full embedding, as a corollary of [11]) and an empirically-supported Angular-Preservation *conjecture*; (2) a measurement with controls (§5); (3) universality across five substrate types (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); and (5) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings.

## 2 Background

**Spectral geometry.** Laplacian eigenmaps [1] and diffusion maps [2] embed a graph via low eigenvectors; the commute-time embedding scales mode  $k$  by  $1/\sqrt{\lambda_k}$ . **Commute-time degeneracy.** von Luxburg et al. [11] proved the commute distance of a large geometric graph degenerates,  $R(i, j) \rightarrow 1/k_i + 1/k_j$  ( $k_i$  the degree of node  $i$ ); we read this backwards as the collapse of the *radial* coordinate to local degree. **Row-normalization.** Ng–Jordan–Weiss [5] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [9] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions, explaining  $k$ -means’ post-normalization success; consistency of spectral clustering is treated in [10]. Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. **Emergent geometry and observer theory.** In the Wolfram model [12, 13, 3] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [14]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in  $\mathbb{H}^2$  at low distortion [4, 8, 6]; we test and reject a hyperbolic reframe in §6.

## 3 The Keep-the-Angle Principle

Let  $G$  be connected on  $n$  nodes with symmetric-normalized Laplacian  $L = I - D^{-1/2}AD^{-1/2}$ , eigenpairs  $(\lambda_k, v_k)$ ,  $0 = \lambda_0 < \lambda_1 \leq \dots$ . The low-mode spectral embedding using the lowest  $m$  nontrivial modes places node  $i$  at  $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$ . Write  $\Psi_i = r_i \hat{u}_i$  with radius  $r_i = \|\Psi_i\|$  and angle  $\hat{u}_i = \Psi_i/\|\Psi_i\|$ .<sup>1</sup>

**Intrinsic dimension and degree notation.** Throughout,  $d$  is the intrinsic dimension of the substrate, and  $k_i$  denotes the degree of node  $i$  (we reserve  $d$  for dimension to avoid the collision  $d$  vs.  $d_i$ ): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices, sphere), and for emergent graphs the value at which the three independent estimators of §4

<sup>1</sup>We call  $\Psi$  “commute-style” rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all  $n - 1$  modes, so that  $\|\Psi_i - \Psi_j\|^2$  equals the resistance  $R(i, j)$  exactly. Our  $\Psi$  is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Theorem 1 and is handled there.

(ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. The regime  $d \geq 2$  refers to this measured intrinsic dimension; §5.4 confirms the  $d = 1$  boundary behaves as predicted.

**Theorem 1** (Radial Degeneracy). *Let  $G_n$  be random geometric graphs of intrinsic dimension  $d \geq 2$  satisfying the conditions of [11], and let  $\Phi_i$  be the full commute-time embedding built from the unnormalized Laplacian, so that  $\|\Phi_i\|^2 = L_{ii}^+$ . As  $n \rightarrow \infty$  the resistance distance degenerates,  $R(i, j) \rightarrow 1/k_i + 1/k_j$ ; since the only per-node content of this additively separable form is the self-term  $L_{ii}^+$ , the radius satisfies  $\|\Phi_i\| \rightarrow 1/\sqrt{k_i}$ . The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density coordinate carrying no geodesic information.*

*Remark* (Truncated, normalized radius). Our experiments use the truncated radius  $r_i = \|\Psi_i\|$  ( $m$  lowest modes of  $L_{\text{sym}}$ ), not  $\|\Phi_i\|$ . Transferring Theorem 1 to this setting requires (i) the normalized analogue of the degeneracy in [11] and (ii) control of the truncation error, and we do not prove either here; we state the truncated statement as an empirical observation. Numerically it holds tightly:  $\rho(r_i, 1/\sqrt{k_i}) = 0.92\text{--}0.99$  and radius-only geodesic preservation  $\rho \leq 0.08$  (§5.5).

**Conjecture 1** (Angular Preservation). *Row-normalization  $\hat{u}_i = \Psi_i/\|\Psi_i\|$  deletes the degenerate density factor of Theorem 1, and the pairwise distances among the resulting directions remain rank-faithful to geodesic distance—Spearman correlation bounded away from zero (empirically  $\approx 0.9$ )—uniformly in the mode-count  $m$  and size  $n$ . A stronger, metric form of the conjecture—that the angular map is bi-Lipschitz to geodesic distance with constants uniform in  $m$  and  $n$ —is consistent with our data but untested; our evidence is rank-based and cannot by itself bound metric distortion.*

*Status.* Empirically strong (§5) but **not proven**. It rests on eigenmap-embedding results plus a heuristic; the honest gap is that Theorem 1 explains why the radius *dies* for  $d \geq 2$ , not by itself why the angle *lives*—indeed the angular fidelity persists ( $\rho \approx 0.88\text{--}0.90$ ) even on quasi-1D graphs where Theorem 1 does not apply, so the angular half is carried by the eigenmap-embedding mechanism, not degeneracy alone. Together these are the **Keep-the-Angle Principle**: keep  $\hat{u}_i$ , discard  $r_i$ .

**Dimension gating.** Theorem 1 is a  $d \geq 2$  phenomenon. In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Theorem 1 correctly does not apply—a prediction, not a failure, confirmed in §5.5.

## 4 Methods

**Estimators.** Ball-growth  $N(r) \sim r^d$ ; spectral dimension from the heat-kernel return probability  $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$ ; effective rank via local-PCA participation ratio. Low inter-estimator spread  $\Rightarrow$  clean manifold, and the agreed value is our measured intrinsic dimension  $d$ . **Angular metric.** Spearman  $\rho$  between embedded pairwise distances and true graph geodesics for: *angle* (unit-normalized low-mode embedding), *random* ( $m$  random modes; control), *magnitude* (radius only), *euclid* (classical 2D MDS). **Uncertainty.** Each  $\rho$  is a Spearman correlation over the  $\binom{a}{2}$  pairs of  $a$  anchor nodes ( $a = 150$  unless noted; 11,175 pairs), with bootstrap 95% CIs over pair resampling (2000 resamples). On the 2-torus ( $n = 2000$ ) the angular basis gives  $\rho = 0.894$ , CI  $[0.888, 0.899]$ , and  $0.897 \pm 0.012$  across 8 independent graph draws; the magnitude control gives  $0.006$   $[0.000, 0.025]$  and the random-mode control  $0.037$   $[0.018, 0.055]$ . **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and a swiss-roll embedding trajectory: a synthetic 2D manifold (spiral angle  $u \in [1.5\pi, 4.5\pi]$ , height  $v \in [0, 20]$ ) carried into  $\mathbb{R}^{12}$  by a random orthogonal map with isotropic Gaussian noise ( $\sigma = 0.03$ ), connected

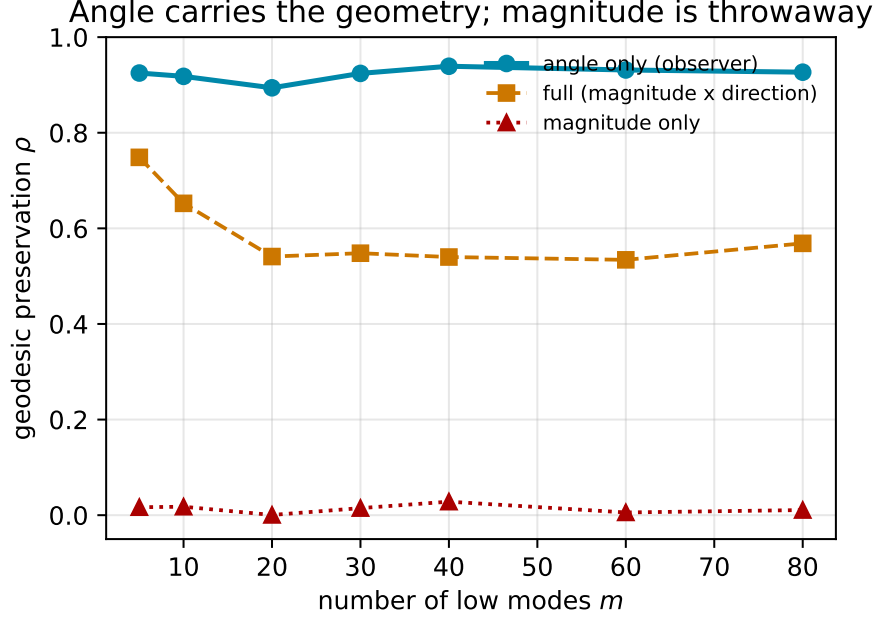


Figure 1: Spearman correlation with graph geodesics on a 2-torus ( $n = 2000$ ) versus mode-count  $m$ : angle carries the geometry, magnitude is throwaway, and the full commute embedding decays with  $m$ .

as a  $k = 10$  nearest-neighbour graph over  $n = 2000$  points—the analogue of embedding a corpus line-by-line onto a low-dimensional semantic manifold and joining nearest lines.

## 5 Results

### 5.1 The core result

Estimators recover ground-truth dimension within  $\approx 0.3$ . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

| $m$ | full (mag. $\times$ dir.) | angle only   | magnitude only |
|-----|---------------------------|--------------|----------------|
| 5   | 0.761                     | <b>0.926</b> | 0.025          |
| 10  | 0.650                     | <b>0.913</b> | 0.035          |
| 20  | 0.550                     | <b>0.891</b> | 0.028          |
| 40  | 0.517                     | <b>0.928</b> | 0.034          |

The angle carries the geometry ( $\approx 0.92$ , flat in  $m$ ); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at  $\rho \approx 0$  (Fig. 2).

### 5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only  $\approx 1.5$ D manifolds (82 manifold-grade rules in a 20k sweep; none at  $d \geq 2.5$ ). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ( $d = 2.07 \pm 0.01$ , spread 0.05, angle- $\rho = 0.82$ ) and by from-scratch evolutionary search ( $\approx 3$  generations). The angular basis holds on genuine 2D emergent space.

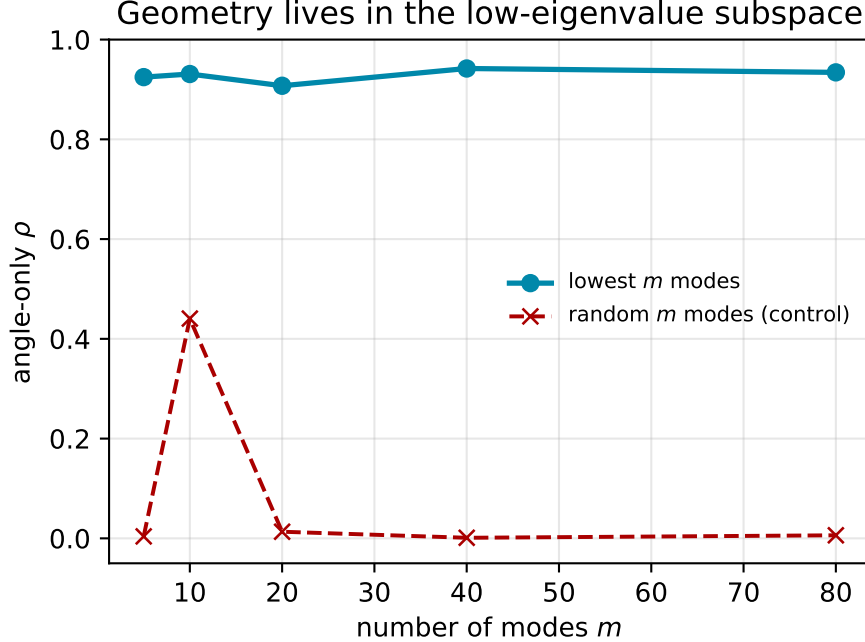


Figure 2: Geometry lives in the low-eigenvalue subspace: lowest- $m$  modes preserve geodesics; a random-mode basis of equal size does not.

### 5.3 Universality across substrates

| substrate                   | meas. $d$ | angle- $\rho$ | random | euclid |
|-----------------------------|-----------|---------------|--------|--------|
| 2-torus (reference)         | 2         | 0.89          | 0.01   | 0.67   |
| king-lattice                | 2         | 0.82          | 0.01   | 0.95   |
| swiss-roll trajectory       | 2         | 0.93          | 0.01   | 0.99   |
| emergent $R_{3D}$ (Wolfram) | 2.07      | 0.82          | 0.01   | 0.81   |
| causal set (Lorentzian)     | 3.9       | 0.57          | 0.01   | 0.55   |
| small-world (destroyed)     | 3.0*      | 0.45          | 0.00   | —      |

Substrates with genuine low- $d$  Riemannian geometry all show the effect, independent of graph origin; the metric fails correctly where geometry is Lorentzian or destroyed. \*For the small-world graph the three dimension estimators do *not* strongly disagree (spread 0.21, comparable to the clean lattice’s 0.22); rather, 3% rewiring inflates the measured dimension from the construction value 2 to  $\approx 3$  and collapses angle- $\rho$  to 0.45. Here the fidelity metric itself—not the spread criterion—detects the destroyed geometry: this is the predicted failure, not an exception.

Two caveats keep the universality claim honest. First, on the two flat substrates (king-lattice, swiss-roll) classical MDS *beats* the angular basis (0.95 and 0.99 vs. 0.82 and 0.93): MDS wins whenever the manifold embeds near-isometrically in low-dimensional Euclidean space, while the angle wins where topology obstructs such an embedding (the torus: 0.89 vs. 0.67) and, unlike MDS, is uniform in  $m$  and  $n$  (§5.5). Universality here means the angular basis never *loses* the geometry on a genuine low- $d$  Riemannian substrate—not that it dominates every baseline on every substrate. Second, the swiss-roll trajectory is the sequential-embedding case: the angular basis is the same whether the manifold comes from hypergraph rewriting or sequential-trajectory data.

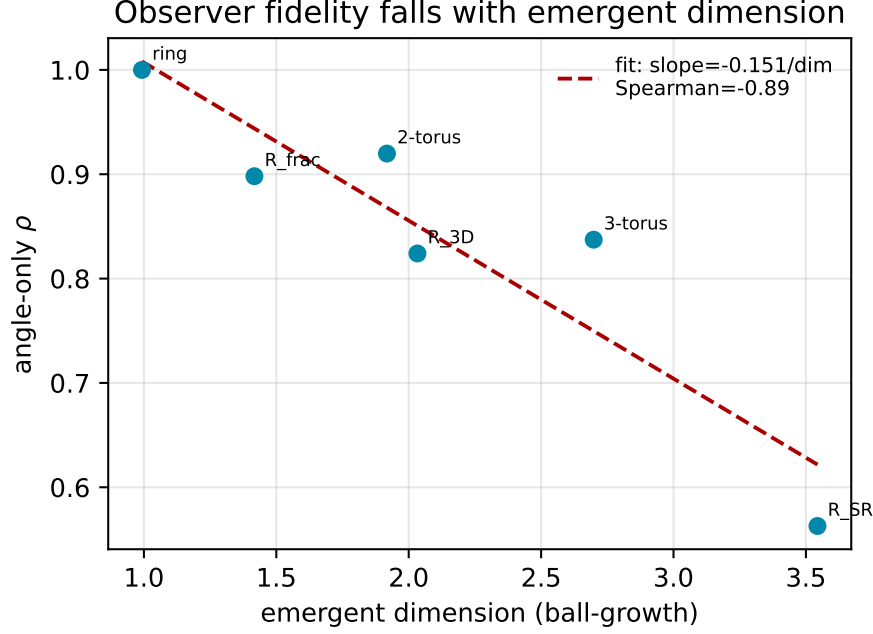


Figure 3: Angular fidelity falls approximately linearly with emergent dimension across ring, tori, and Wolfram-model rules (slope  $-0.157 \pm 0.028$ ,  $R^2 = 0.64$  over 20 manifolds). Per-point error bars from pair-resampling are left to future work.

#### 5.4 The scaling law

Across 20 independent emergent manifolds ( $d \approx 1\text{--}3.6$ ),

$$\text{angle-}\rho \approx 1.09 - 0.157 d, \quad \text{slope } -0.157 \pm 0.028, \quad R^2 = 0.64, \quad d \in [1.0, 3.6].$$

(Spearman  $-0.881$  between  $\rho$  and  $d$  serves as a distribution-free monotonicity check; the linear fit itself should be judged by its slope uncertainty and  $R^2$ .) The fit is *empirical and local*:  $\rho$  is a rank correlation bounded by 1, so the linear form holds only within the measured range—the intercept 1.09 is an extrapolation, not a literal  $d \rightarrow 0$  fidelity, and the slope may vary with substrate roughness. Within range, fidelity is highest at low dimension and softens toward  $\approx 0.56$  at  $d \approx 3.4$  (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

#### 5.5 Verifying Theorem 1 and Conjecture 1

angle- $\rho$  is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ( $\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.92\text{--}0.99$ ; radius-only  $\rho \leq 0.08$ ). Theorem 1 holds cleanly at  $d = 3$  ( $\rho(R, 1/k_i + 1/k_j) \rightarrow 0.98$ );  $d = 2$  is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 1: in an  $m$ -sweep at  $n = 4000$ , angle- $\rho$  is flat  $\approx 0.93$  across  $m = 5\text{--}80$  while the full embedding decays  $0.773 \rightarrow 0.437$ .

## 6 Honest Negatives

**Not secretly hyperbolic (rejected).** With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the

arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw  $-0.98$  against update density that deflates to a tautology (update-count equals degree,  $\rho = +1.00$ ; hubs are negatively curved); controlling for degree, the independent signal is partial- $\rho = -0.14$ . This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

## 7 Discussion and Conclusion

**The primary result is basis-level and substrate-agnostic.** The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a geodesic justification complementary to the cluster-separation theory of [9], unifies it with the scale-invariant direction of KV-cache compression [7], and holds *universally* across substrates with genuine low- $d$  Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-rewriting and sequential-trajectory substrates is evidence that scale-invariant angular coarse-graining is a general mechanism by which relational data carries geometric structure.

**One interpretation is observer-theoretic.** In Wolfram’s observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making “the observer’s basis” a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy-responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving Conjecture 1 (in its rank form, and ideally the bi-Lipschitz strengthening); extending Theorem 1 to the truncated, normalized radius actually used in practice; synthesizing clean manifolds above  $d \approx 2$ ; and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude: the basis-level fact is the contribution; the observer interpretation is where we leave the door open.

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